

User Simulation vs. Real Interaction. Our "casual" prompts are proxies for real user feedback. However, these responses are too limited to definitively translate our results to LLM-user interaction.

Prompt Sensitivity. LLM responses are known to be highly sensitive to even small variations in prompt wording [Zhuo et al. \(2024\)](#). Slight differences in phrasing could greatly alter our results.

Disagreement Sample Bias We randomly sample disagreement pairs from a pool of responses. As a result, less performant model responses are more likely to be paired with highly performant model responses. This introduces a bias that may partially confound our persuasion percentage.

Acknowledgments

We'd like to thank Taiming Lu, Ziang Xiao for insightful guidance in creating this paper. The authors were supported by ONR grant (N00014-24-1-2089).

References

- DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bing-Li Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Daya Guo, Dejian Yang, Deli Chen, Dong-Li Ji, Erhang Li, Fangyun Lin, Fucong Dai, and 179 others. 2024. [Deepseek-v3 technical report](#). *ArXiv*, abs/2412.19437.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2024. [Improving factuality and reasoning in language models through multiagent debate](#). In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 11733–11763. PMLR.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, Anirudh Goyal, Anthony S. Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, and 510 others. 2024. [The llama 3 herd of models](#). *ArXiv*, abs/2407.21783.
- Andrew Estornell and Yang Liu. 2025. Multi-llm debate: framework, principals, and interventions. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, NIPS '24, Red Hook, NY, USA. Curran Associates Inc.
- Aaron Fanous, Jacob Goldberg, Ank A. Agarwal, Joanna Lin, Anson Zhou, Roxana Daneshjou, and Sanmi Koyejo. 2025. [Syceval: Evaluating llm syco-phancy](#). *Preprint*, arXiv:2502.08177.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. [Measuring massive multitask language understanding](#). In *International Conference on Learning Representations*.
- Chi Hu, Yuan Ge, Xiangnan Ma, Hang Cao, Qiang Li, Yonghua Yang, Tong Xiao, and Jingbo Zhu. 2024. [RankPrompt: Step-by-step comparisons make language models better reasoners](#). In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13524–13536, Torino, Italia. ELRA and ICCL.
- Zhengyu Hu, Linxin Song, Jieyu Zhang, Zheyuan Xiao, Tianfu Wang, Zhengyu Chen, Nicholas Jing Yuan, Jianxun Lian, Kaize Ding, and Hui Xiong. 2025. [Explaining length bias in llm-based preference evaluations](#). *Preprint*, arXiv:2407.01085.
- Takeshi Kojima, Shixiang (Shane) Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. [Large language models are zero-shot reasoners](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 22199–22213. Curran Associates, Inc.
- Satyapriya Krishna, Chirag Agarwal, and Himabindu Lakkaraju. 2024. [Understanding the effects of iterative prompting on truthfulness](#). In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 25583–25602. PMLR.
- Philippe Laban, Lidiya Murakhovska, Caiming Xiong, and Chien-Sheng Wu. 2024. [Are you sure? challenging llms leads to performance drops in the flipflop experiment](#). *Preprint*, arXiv:2311.08596.
- Harrison Lee, Samrat Phatale, Hassan Mansoor, Thomas Mesnard, Johan Ferret, Kellie Ren Lu, Colton Bishop, Ethan Hall, Victor Carbune, Abhinav Rastogi, and Sushant Prakash. 2024. [RLAIF vs. RLHF: Scaling reinforcement learning from human feedback with AI feedback](#). In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 26874–26901. PMLR.
- Dawei Li, Bohan Jiang, Liangjie Huang, Alimohammad Beigi, Chengshuai Zhao, Zhen Tan, Amrita Bhat-tacharjee, Yuxuan Jiang, Canyu Chen, Tianhao Wu, Kai Shu, Lu Cheng, and Huan Liu. 2025. [From generation to judgment: Opportunities and challenges of llm-as-a-judge](#). *Preprint*, arXiv:2411.16594.
- Haitao Li, Qian Dong, Junjie Chen, Huixue Su, Yujia Zhou, Qingyao Ai, Ziyi Ye, and Yiqun Liu. 2024. [Llms-as-judges: A comprehensive survey on llm-based evaluation methods](#). *Preprint*, arXiv:2412.05579.
- Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. 2024. [Encouraging divergent thinking in large language models through multi-agent debate](#). *Preprint*, arXiv:2305.19118.

- Jian Liu, Leyang Cui, Hanmeng Liu, Dandan Huang, Yile Wang, and Yue Zhang. 2020. [Logiqa: A challenge dataset for machine reading comprehension with logical reasoning](#). In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, pages 3622–3628. International Joint Conferences on Artificial Intelligence Organization. Main track.
- Joshua Liu, Aarav Jain, Soham Takuri, Srihan Vege, Aslihan Akalin, Kevin Zhu, Sean O’Brien, and Vasu Sharma. 2025. [Truth decay: Quantifying multi-turn sycophancy in language models](#). *Preprint*, arXiv:2503.11656.
- Meta AI. 2024. The Llama 4 herd: The beginning of a new era of natively multimodal AI innovation. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>. Accessed: 2025-08-22.
- Ankit Pal, Logesh Kumar Umapathi, and Malaikanan Sankarasubbu. 2022. Medmcqa: A large-scale multi-subject multi-choice dataset for medical domain question answering. In *Conference on health, inference, and learning*, pages 248–260. PMLR.
- Ethan Perez, Sam Ringer, Kamile Lukosiute, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, Andy Jones, Anna Chen, Benjamin Mann, Brian Israel, Bryan Seethor, Cameron McKinnon, Christopher Olah, Da Yan, Daniela Amodei, and 44 others. 2023. [Discovering language model behaviors with model-written evaluations](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13387–13434, Toronto, Canada. Association for Computational Linguistics.
- Ethan Perez, Sam Ringer, Kamilė Lukošiušė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, Andy Jones, Anna Chen, Ben Mann, Brian Israel, Bryan Seethor, Cameron McKinnon, Christopher Olah, Da Yan, Daniela Amodei, and 44 others. 2022. [Discovering language model behaviors with model-written evaluations](#). *Preprint*, arXiv:2212.09251.
- Keita Saito, Akifumi Wachi, Koki Wataoka, and Youhei Akimoto. 2023. [Verbosity bias in preference labeling by large language models](#). *Preprint*, arXiv:2310.10076.
- Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Esin DURMUS, Zac Hatfield-Dodds, Scott R Johnston, Shauna M Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. 2024. [Towards understanding sycophancy in language models](#). In *The Twelfth International Conference on Learning Representations*.
- Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. [CommonsenseQA: A question answering challenge targeting commonsense knowledge](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4149–4158, Minneapolis, Minnesota. Association for Computational Linguistics.
- Arne Tillmann. 2025. [Literature review of multi-agent debate for problem-solving](#). *Preprint*, arXiv:2506.00066.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. 2023. [Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting](#). *Preprint*, arXiv:2305.04388.
- Boshi Wang, Xiang Yue, and Huan Sun. 2023. [Can chatgpt defend its belief in truth? evaluating llm reasoning via debate](#). *Preprint*, arXiv:2305.13160.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS ’22*, Red Hook, NY, USA. Curran Associates Inc.
- Qingjie Zhang, Han Qiu, Di Wang, Haoting Qian, Yiming Li, Tianwei Zhang, and Minlie Huang. 2024. [Understanding the dark side of llms’ intrinsic self-correction](#). *Preprint*, arXiv:2412.14959.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging llm-as-a-judge with mt-bench and chatbot arena](#). *Preprint*, arXiv:2306.05685.
- Jingming Zhuo, Songyang Zhang, Xinyu Fang, Haodong Duan, Dahua Lin, and Kai Chen. 2024. [ProSA: Assessing and understanding the prompt sensitivity of LLMs](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1950–1976, Miami, Florida, USA. Association for Computational Linguistics.

A Model Snapshots, API provider, and Cost of LLMs

Model info / snapshot	API Provider
Deepseek V3 DeepSeek-AI et al. (2024)	Together.ai
gpt-4.1-2025-04-14	OpenAI
gpt-4.1-mini-2025-04-14	OpenAI
gpt-4.1-nano-2025-04-14	OpenAI
gpt-4o-mini-2024-07-18	OpenAI
Llama-3.3-70B-Instruct-Turbo Dubey et al. (2024)	Together.ai
Llama-4-Maverick-17B-128E-Instruct-FP8 Meta AI (2024)	Together.ai
Llama-4-Scout-17B-16E-Instruct Meta AI (2024)	Groq

Table 9: Used language model info, including API providers. The total API usage for this study, including preliminary experimental runs, amounted to approximately \$100.

B Zero-shot CoT LLM accuracies

Referring back to [subsection 3.2](#), this heatmap shows LLM accuracy across different MCQ datasets. The ARC Challenge, ARC Easy and SciQ had very high accuracy among models, with the supermajority achieving accuracy of over 95%. These dataset results were excluded as it offered insufficient number of disagreement pairs.

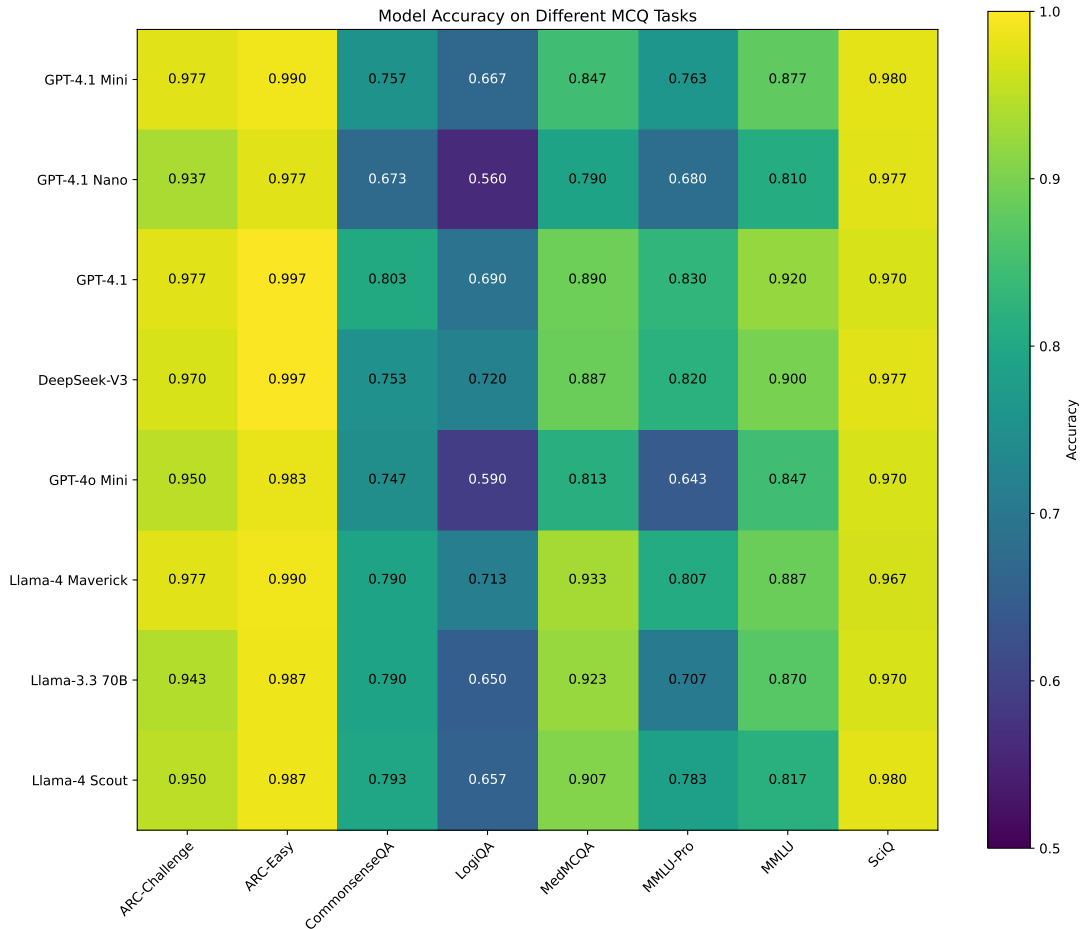


Figure 3: Heatmap of zero-shot Chain-of-Thought (CoT) accuracies for each LLM across the initial set of MCQ datasets. Datasets where most models achieved over 95% accuracy (e.g., ARC, SciQ) were excluded from our main analysis due to an insufficient number of disagreement pairs.